Your Solved Arrays Programs — All 10

Q1. Find Maximum Element

java

```
import java.util.*;

public class Main {

    public static void print(int arr[]) {

        for(int i= 0; i<arr.length;i++) {

            System.out.print(arr[i] +" ");

        }

        System.out.println();

    }

    public static int max(int arr[]) {

       int max = Integer.MIN_VALUE;

        for(int i = 0;i<arr.length; i++) {

            if(arr[i]>max) {

                max = arr[i];

            }

        }

     return max;

    }

    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the size of the array:");

        int n = sc.nextInt();

        int arr[] = new int[n];

        for(int i = 0; i<n; i++) {

            arr[i] = sc.nextInt();

        }

        print(arr);

        System.out.println(max(arr));

    }

}
```

Output:

3 7 1 9 4

Maximum = 9

Q2. Find Minimum Element

java

```
import java.util.*;

public class Main {

    public static void print(int arr[]) {

        for(int i= 0; i<arr.length;i++) {

            System.out.print(arr[i] +" ");

        }

        System.out.println();
```

```
    }
    public static int min(int arr[]) {
      int min = Integer.MAX_VALUE;
       for(int i = 0;i<arr.length; i++) {
           if(arr[i]<min) {
               min = arr[i];
           }
       }
     return min;
    }
    public static void main(String[] args) {
       Scanner sc = new Scanner(System.in);
       System.out.println("Enter the size of the array:");
       int n = sc.nextInt();
       int arr[] = new int[n];
       for(int i = 0; i<n; i++) {
           arr[i] = sc.nextInt();
       }
       print(arr);
       System.out.println(min(arr));
    }
}
```

Output:

3 7 1 9 4

Minimum = 1

Q3. Sum and Average

java

```
import java.util.*;

public class Main {

    public static void print(int arr[]) {

        for(int i= 0; i<arr.length;i++) {

            System.out.print(arr[i] +" ");

        }

        System.out.println();

    }

    public static void avg(int arr[]) {

      int sum = 0;

      double avg;

      for(int i = 0;i<arr.length; i++) {

          sum = sum + arr[i];

      }

      System.out.println("Sum = " + sum);

      avg = (double)sum/arr.length;

      System.out.println("Average = " + avg);

    }
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the size of the array:");
        int n = sc.nextInt();
        int arr[] = new int[n];
        for(int i = 0; i<n; i++) {
            arr[i] = sc.nextInt();
        }
        print(arr);
        avg(arr);
    }
}
```

Output:

3 7 1 9 4

Sum = 24

Average = 4.8

Q4. Reverse Array

java

```
import java.util.*;
class Main {
    public static void print(int arr[]) {
        for(int i = 0; i<arr.length;i++) {
            System.out.print(arr[i] +" ");
```

```
    }
    System.out.println();
}
public static void reverse(int arr[]) {
    int st = 0;
    int end = arr.length -1;
    while(st<end) {
        int temp = arr[st];
        arr[st] = arr[end];
        arr[end] = temp;
        st++;
        end--;
    }
}
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.println("Enter the size of array:");
    int n = sc.nextInt();
    int arr[] = new int[n];
    for(int i = 0; i<n; i++) {
        arr[i] = sc.nextInt();
    }
    print(arr);
```

```
        reverse(arr);
        print(arr);
    }
}
```

Output:

Before: 1 2 3 4 5

After: 5 4 3 2 1

Q5. Palindrome Check

java

```
import java.util.*;
class Main {
    public static void print(int arr[]) {
        for(int i = 0; i<arr.length;i++) {
            System.out.print(arr[i] +" ");
        }
        System.out.println();
    }
    public static void reverse(int arr[]) {
        int st = 0;
        int end = arr.length -1;
        while(st<end) {
            if(arr[st]!= arr[end]) {
                System.out.println("Array is Not Palindrome");
```

```
            return;
        }
        st++;
        end--;
    }
    System.out.println("Array is Palindrome");
  }
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.println("Enter the size of array:");
    int n = sc.nextInt();
    int arr[] = new int[n];
    for(int i = 0; i<n; i++) {
        arr[i] = sc.nextInt();
    }
    print(arr);
    reverse(arr);
  }
}
```

Output:

1 2 3 2 1 → Array is Palindrome

1 2 3 4 → Array is Not Palindrome

Q6. Linear Search

java

```
import java.util.*;

class Main {

   public static void print(int arr[]) {

      for(int i = 0; i<arr.length;i++) {

         System.out.print(arr[i] +" ");

      }

      System.out.println();

   }

   public static int search(int arr[], int key) {

      for(int i = 0; i<arr.length;i++) {

         if(arr[i] == key) {

            return i;

         }

      }

      return -1;

   }

   public static void main(String[] args) {

      Scanner sc = new Scanner(System.in);

      System.out.println("Enter the size of array:");

      int n = sc.nextInt();

      int arr[] = new int[n];
```

```
        for(int i = 0; i<n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.print("Enter the search Element:");
        int key = sc.nextInt();
        print(arr);
        System.out.println(search(arr, key));
    }
}
```

Output:

Search 7 → Found at index 1

Search 5 → -1 (Not Found)

Q7. Selection Sort

java

```
import java.util.*;
class Main {
    public static void print(int arr[]) {
        for(int i = 0; i<arr.length;i++) {
            System.out.print(arr[i] +" ");
        }
        System.out.println();
    }
    public static void sort(int arr[]) {
```

```
        for(int i = 0; i<arr.length;i++) {
          for(int j = i + 1;j<arr.length;j++) {
            if(arr[i]>arr[j]) {
              int temp = arr[i];
              arr[i] = arr[j];
              arr[j] = temp;
            }
          }
        }
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int n = sc.nextInt();
        int arr[] = new int[n];
        for(int i = 0; i<n; i++) {
           arr[i] = sc.nextInt();
        }
        print(arr);
        sort(arr);
        print(arr);
    }
}
```

Output:

Before: 5 3 8 1 4

After: 1 3 4 5 8

Q8. Find Duplicates

java

```
import java.util.*;

class Main {

   public static void print(int arr[]) {

      for(int i = 0; i<arr.length;i++) {

         System.out.print(arr[i] +" ");

      }

      System.out.println();

   }

   public static void duplicates(int arr[]) {

      for(int i = 0; i<arr.length;i++) {

        for(int j = i + 1;j<arr.length;j++) {

           if(arr[i] == arr[j]) {

             System.out.print(arr[i] +" ");

           }

        }

      }

   }

   public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int n = sc.nextInt();
        int arr[] = new int[n];
        for(int i = 0; i<n; i++) {
            arr[i] = sc.nextInt();
        }
        print(arr);
        duplicates(arr);
    }
}
```

Output:

1 2 3 2 4 1

Duplicates: 1 2

Q9. Second Largest Element

java

```
import java.util.*;
class Main {
    public static void print(int arr[]) {
        for(int i = 0; i<arr.length;i++) {
            System.out.print(arr[i] +" ");
        }
        System.out.println();
```

```
}

public static void secondLargest(int arr[]) {

   for(int i = 0; i<arr.length; i++) {

      for(int j = i + 1; j<arr.length; j++) {

         if(arr[i]>arr[j]) {

            int temp = arr[i];

            arr[i] = arr[j];

            arr[j] = temp;

         }

      }

   }

   System.out.println("Second Largest = " + arr[arr.length - 2]);

}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.println("Enter the size of array:");

    int n = sc.nextInt();

    int arr[] = new int[n];

    for(int i = 0; i<n; i++) {

        arr[i] = sc.nextInt();

    }

    print(arr);

    secondLargest(arr);
```

```
    }
}
```

Output:

3 7 1 9 4

Second Largest = 7

Q10. Move Zeros to End

java

```java
import java.util.*;
class Main {
    public static void print(int arr[]) {
        for(int i = 0; i<arr.length; i++) {
            System.out.print(arr[i] +" ");
        }
        System.out.println();
    }
    public static void lastz(int arr[]) {
        int temp[] = new int[arr.length];
        int j = 0;
        for(int i = 0; i<arr.length; i++) {
            if(arr[i] != 0) {
                temp[j++] = arr[i];
            }
        }
```

```
        while(j<arr.length) {
            temp[j++] = 0;
        }
        for(int i = 0; i<arr.length; i++) {
            arr[i] = temp[i];
        }
    }
    public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.println("Enter the size of array:");
      int n = sc.nextInt();
      int arr[] = new int[n];
      for(int i = 0; i<n; i++) {
          arr[i] = sc.nextInt();
      }
      lastz(arr);
      print(arr);
    }
}
```

Output:

Before: 4 0 5 0 1 9 0

After: 4 5 1 9 0 0 0